Smart Leave Management System

Track 4 – Workforce & Enterprise | Problem Statement 25

Abstract

Managing employee leave through emails, spreadsheets, and verbal approvals leads to delays, inconsistent policy enforcement, inaccurate balance tracking, and poor visibility into team availability. This project presents Smart Leave Management, a full-stack web platform that lets employees apply for leave and managers approve or reject requests through a structured, transparent workflow.

The system supports role-based access for employees, managers, and HR administrators. Employees can view their leave balances by leave type (casual, sick, earned, and so on), submit requests with dates and reasons, and track status in real time. Managers receive pending requests, review them alongside team availability, and approve or reject with remarks. The backend applies business rules automatically: it validates date ranges, prevents overlapping requests, checks available balances, and deducts or restores balances on approval, rejection, or cancellation. A complete leave history is maintained for every employee, and HR can configure leave types and generate reports.

The application is built with React.js on the frontend and Java Spring Boot on the backend, exposing REST APIs secured with Spring Security and role-based authorization. Data is persisted in MySQL/PostgreSQL through Spring Data JPA/Hibernate, with relationships among employees, departments, leave types, balances, and leave requests.

As the intelligent component, a supporting Python module analyzes historical leave data to identify patterns such as peak leave periods, frequent short-notice absences, and department-level trends. These insights are surfaced on the manager and HR dashboards to support better workforce planning.

Keywords: Leave Management, Approval Workflow, Spring Boot, REST API, Role-Based Access, Leave Pattern Analysis

Technology Requirements

Area	Expected Technology
Frontend	React.js or Angular
Backend	Java with Spring Boot, Spring MVC / REST APIs
Persistence	MySQL or PostgreSQL using Spring Data JPA / Hibernate
Security	Spring Security / role-based access where applicable
Python	Limited to AI/ML, NLP, prediction, analytics, recommendation, or intelligent processing

Important implementation rule: Java Spring Boot must be the primary application backend. Python must not replace the Java backend and may be used only as a supporting intelligence/analytics component.

Source: CODESPARK HACKATHON 2026 Problem Statement Compendium.